THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 5: **Parallel and Intersecting Lines** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 58	$3x + a - b = 180$.
2	**(a)** 112°	Co-interior angles are supplementary.
3	**(c)** 60°	$x = k(90 - x)$.
4	**(c)** 70°	Supplement first, then halve.
5	**(b)** 35°	Bisector splits equally.
6	**(a)** parallel	Check the angle relation.
7	**(a)** 60°	Total 180°.
8	**(c)** 56	$3x + a - b = 180$.
9	**(c)** 93°	Co-interior angles are supplementary.
10	**(a)** 75°	$x = k(90 - x)$.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$x = (180 - x) - 20$ (1); $2x = 160$, $x = 80^\circ$ (1).	1+1
12	Vertically opposite angles are equal (1); $2x = 30$, $x = 15$ (1).	1+1
13	Compare 75° and 75° (1); Equal corresponding angles imply parallel lines (1).	1+1
14	$x = (180 - x) - 40$ (1); $2x = 140$, $x = 70^\circ$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Co-interior angles sum to 180° (1); $5x = 155$, $x = 31$ (1); Angles 72°, 108° (1).	1+1+1
16	$3x + 6 = 180$; $x = 58$ (1); Angles 73° and 107° (1); Each equals its vertically opposite (1).	1+1+1
17	$x = 2(90 - x) + 6$ (1); $3x = 186$, $x = 62^\circ$ (1); Complement 28° (1).	1+1+1
	OR: Co-interior angles sum to 180° (1); $5x = 155$, $x = 31$ (1); Angles 67°, 113° (1).	1+1+1
18	$3x + 24 = 180$; $x = 52$ (1); Angles 82° and 98° (1); Each equals its vertically opposite (1).	1+1+1
	OR: $x = 2(90 - x) + 15$ (1); $3x = 195$, $x = 65^\circ$ (1); Complement 25° (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Alternate interior equal: $2x + a = 3x - b$ (1); $x = 30$ (1); $\angle 1 = \angle 2 = 70^\circ$ (1); $\angle 3 = 180^\circ - 70^\circ = 110^\circ$ (1); Reasons (1).	1+1+1+1+1
	OR: $6x + 30 = 360$ (1); $x = 55$ (1); Angles $55^\circ, 110^\circ, 165^\circ, 30^\circ$ (1); Classify each (2).	1+1+1+2
20	Alternate interior equal: $2x + a = 3x - b$ (1); $x = 25$ (1); $\angle 1 = \angle 2 = 65^\circ$ (1); $\angle 3 = 180^\circ - 65^\circ = 115^\circ$ (1); Reasons (1).	1+1+1+1+1
	OR: $6x + 60 = 360$ (1); $x = 50$ (1); Angles $50^\circ, 100^\circ, 150^\circ, 60^\circ$ (1); Classify each (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 5: **Parallel and Intersecting Lines** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 60°	Supplement first, then halve.
2	**(c)** 55°	Bisector splits equally.
3	**(c)** not parallel	Check the angle relation.
4	**(b)** 72°	Total 180°.
5	**(d)** 63	$3x + a - b = 180$.
6	**(c)** 81°	Co-interior angles are supplementary.
7	**(c)** 80°	$x = k(90 - x)$.
8	**(d)** 40°	Supplement first, then halve.
9	**(d)** 65°	Bisector splits equally.
10	**(a)** parallel	Check the angle relation.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Vertically opposite angles are equal (1); $2x = 20$, $x = 10$ (1).	1+1
12	Compare 110° and 105° (1); Unequal, so not parallel (1).	1+1
13	$x = (180 - x) - 60$ (1); $2x = 120$, $x = 60^\circ$ (1).	1+1
14	Vertically opposite angles are equal (1); $2x = 52$, $x = 26$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Co-interior angles sum to 180° (1); $5x = 155$, $x = 31$ (1); Angles 77°, 103° (1).	1+1+1
16	$3x + 9 = 180$; $x = 57$ (1); Angles 69° and 111° (1); Each equals its vertically opposite (1).	1+1+1
17	$x = 2(90 - x) + 30$ (1); $3x = 210$, $x = 70^\circ$ (1); Complement 20° (1).	1+1+1
	OR: Co-interior angles sum to 180° (1); $5x = 135$, $x = 27$ (1); Angles 74°, 106° (1).	1+1+1
18	$3x + 12 = 180$; $x = 56$ (1); Angles 80° and 100° (1); Each equals its vertically opposite (1).	1+1+1
	OR: $x = 2(90 - x) + 45$ (1); $3x = 225$, $x = 75^\circ$ (1); Complement 15° (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Alternate interior equal: $2x + a = 3x - b$ (1); $x = 30$ (1); $\angle 1 = \angle 2 = 65^\circ$ (1); $\angle 3 = 180^\circ - 65^\circ = 115^\circ$ (1); Reasons (1).	1+1+1+1+1
	OR: $6x + 90 = 360$ (1); $x = 45$ (1); Angles $45^\circ, 90^\circ, 135^\circ, 90^\circ$ (1); Classify each (2).	1+1+1+2
20	Alternate interior equal: $2x + a = 3x - b$ (1); $x = 35$ (1); $\angle 1 = \angle 2 = 90^\circ$ (1); $\angle 3 = 180^\circ - 90^\circ = 90^\circ$ (1); Reasons (1).	1+1+1+1+1
	OR: $6x + 120 = 360$ (1); $x = 40$ (1); Angles $40^\circ, 80^\circ, 120^\circ, 120^\circ$ (1); Classify each (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 5: **Parallel and Intersecting Lines** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 30°	Total 180°.
2	**(b)** 57	$3x + a - b = 180$.
3	**(a)** 57°	Co-interior angles are supplementary.
4	**(c)** 45°	$x = k(90 - x)$.
5	**(b)** 30°	Supplement first, then halve.
6	**(d)** 75°	Bisector splits equally.
7	**(a)** 3	Check the angle relation.
8	**(d)** 80°	Total 180°.
9	**(a)** 81°	$x = k(90 - x)$.
10	**(d)** 85°	$x = k(90 - x)$.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Compare 62° and 62° (1); Equal corresponding angles imply parallel lines (1).	1+1
12	$x = (180 - x) - 100$ (1); $2x = 80$, $x = 40^\circ$ (1).	1+1
13	Vertically opposite angles are equal (1); $2x = 38$, $x = 19$ (1).	1+1
14	Compare 48° and 84° (1); Unequal, so not parallel (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Co-interior angles sum to 180° (1); $5x = 150$, $x = 30$ (1); Angles 65°, 115° (1).	1+1+1
16	$3x + 3 = 180$; $x = 59$ (1); Angles 68° and 112° (1); Each equals its vertically opposite (1).	1+1+1
17	$x = 2(90 - x) + 12$ (1); $3x = 192$, $x = 64^\circ$ (1); Complement 26° (1).	1+1+1
	OR: Co-interior angles sum to 180° (1); $5x = 140$, $x = 28$ (1); Angles 66°, 114° (1).	1+1+1
18	$3x + 18 = 180$; $x = 54$ (1); Angles 75° and 105° (1); Each equals its vertically opposite (1).	1+1+1
	OR: $x = 2(90 - x) + 24$ (1); $3x = 204$, $x = 68^\circ$ (1); Complement 22° (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Alternate interior equal: $2x + a = 3x - b$ (1); $x = 30$ (1); $\angle 1 = \angle 2 = 68^\circ$ (1); $\angle 3 = 180^\circ - 68^\circ = 112^\circ$ (1); Reasons (1).	1+1+1+1+1
	OR: $6x + 150 = 360$ (1); $x = 35$ (1); Angles $35^\circ, 70^\circ, 105^\circ, 150^\circ$ (1); Classify each (2).	1+1+1+2
20	Alternate interior equal: $2x + a = 3x - b$ (1); $x = 30$ (1); $\angle 1 = \angle 2 = 72^\circ$ (1); $\angle 3 = 180^\circ - 72^\circ = 108^\circ$ (1); Reasons (1).	1+1+1+1+1
	OR: $6x + 36 = 360$ (1); $x = 54$ (1); Angles $54^\circ, 108^\circ, 162^\circ, 36^\circ$ (1); Classify each (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.